Characterizing the Microstructure and Environmentally Assisted Cracking (EAC) of Aluminum 2219 (Compromised)

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This Phase 1 of the test program is as follows:

Determine the Environmentally Assisted Cracking (EAC) susceptibility and threshold performance of compromised (Segregated, with banded and clustered AlCu intermetallics) Aluminum 2219 alloy with focus on ambient condensation environments.

UVa shall not share any data on hardware or materials outside the project. This includes any observations, conclusions, analysis, or reports.

NASA provides: Test coupons of the original lot of Al 2219 as used on the manufacture of various bay segments of the NASA Gateway HALO module as per the recommended spreadsheet test plan provided. If more are required, they will be provided as needed. Test coupons for the new lot will be provided no later than February 28, 2026. NASA KSC personnel with pertinent expertise will be provided to UVa to expedite EAC testing timeline and to resolve any testing anomalies or course corrections.

Phase 1: Evaluate the EAC kinetics of Aluminum 2219 (compromised) in relevant environments.

Objectives:
1) Contractor shall determine the resolution limit of Al 2219 in a dry nitrogen environment to provide an understanding of crack tip plasticity effects of loading as an artifact of the crack monitoring technique.
2) Contractor shall run EAC experiments of Aluminum 2219 coupons in more susceptible environments (3.5% NaCl full immersion) in order to conduct crack tip electron back scatter detection (EBSD).
3) Contractor shall evaluate EAC kinetics and Aluminum 2219 EAC susceptibility on coupons pre-exposed to relevant environments (i.e. - 1x condensate ersatz, but to be determined based on NASA team consensus) followed by rising stress intensity (K) Linear Elastic Fracture Mechanics (LEFM), which may include exposure to various relative humidities (to be determined by NASA team consensus). Minimum of 5, maximum of 10 coupons will be tested. After minimum of 5 coupons are tested, contractor shall coordinate results with NASA to determine if additional coupon testing is required.
4) Contractor shall evaluate loading rate sensitivity of a subset Aluminum 2219 coupons (K-hold) subjected to EAC testing in Objective 3.
5) Contractor shall conduct fractography on all coupons from Objectives 1-4.
6) Contractor shall determine mechanism for EAC in this particular Al 2219 from EAC experiments conducted from Objectives 1-5.
7) Contractor shall periodically provide advisory role on fracture mechanics modeling techniques for the purpose of informing life prediction models (expected 3 times per month) from January 2026 through the end of the period of performance (18 months).
8) Contractor shall provide advisory role on root cause and fault tree closure (if needed) from January 2026 through the end of the period of performance (18 months).
9) OPTION: Contractor shall travel to various locations to support a HALO Module Corrosion Tiger Team Investigation TIM at the request of the Government.

Ground Rules and Assumptions:
1) There may be a need for one domestic trip for 1 person.

2) There may be an option for one international trip for 1 person to Thales Alenia or a sub-tier vendor. The cost associated for these trips, if required, will be provided by the Government. This milestone will only be paid if the contractor is required to travel at the request of the Government.
3) NASA will pay for NASA subject matter expert's time and travel internally and no costs associated with this support are to be included in this proposal.
4) UVa will provide two test rigs for simultaneous use by UVa and NASA experts to expedite the testing on a Non-Interference basis with existing PrK testing activites.
5) Period of performance will be through 7/30/2027 (if needed). The advisory role goes through July 2027, therefore the advisory role performed under this contract will be from February 2026 through the end of the period of performance. If the Government does not require advisory support for the entire period, a portion of the support may be descoped and the final payment milestone adjusted accordingly.

Deliverables:

1) Test plan to meet objectives 1-6, agreed to by NASA.
2) Contractor shall provide incremental status reports: weekly written statuses via email and weekly tag-ups
3) Contractor shall participate in meetings as advisory role with option to travel (1x Domestic for 1 person and Option for 1x International for 1 person).
4) Contractor shall furnish a final report summarizing Phase 1-4 findings and conclusions.

Payment Milestones:

The Contractor shall propose a schedule and reporting cycle. Payment of each milestone will be made after successful completion and COR approval based on the following table:

	Deliverable Summary/Payment Milestones	**Payment %**	**Date**
1.	Contract status every week via email or telecon depending on the nature of the topics.	N/A	Weekly
2.	Test plan to meet objectives 1-6.	20%	ATP + 1.5 months
3.	Completion of objectives 1-6.	50%	ATP + 3.75 months
4.	Final report summarizing test data from objectives 1-6 and summarizing Phase 1-4 findings and conclusions.	20%	ATP + 4 months
5.	Final email report summarizing Advisory support provided.	10%	9/30/2026
6.	OPTION: Completion of one International trip	OFI	TBD

Period of Performance:

The Period of Performance is from 2/28/26 – 12/31/27 (or earlier if tasks are completed).